

CPReady: Development and Evaluation of an ESP32-Based Integrated Supplementary Cardiopulmonary Resuscitation (CPR) Training Device and Mobile Application with Real-Time Objective Feedback and Progress Tracking for STEM-Health Allied Students

Bacergo¹, Balugo², Beltrano³, Bucud⁴, Caraballos⁵, Carranza⁶, Dilig⁷

Bataan National High School – Senior High School

*nycabacergo@gmail.com¹, charlenemargaretteibalugo², cjb.cyrene@gmail.com³,
alsonbucud@gmail.com⁴, caraballosmhovie@gmail.com⁵, keithaudrey423@gmail.com⁶,
diligayeshamhiel@gmail.com⁷*

INTRODUCTION

In 2025, Johnette Wilmot successfully performed cardiopulmonary resuscitation (CPR) on a 15-year-old who collapsed from cardiac arrest, using skills she had learned in high school more than four decades earlier (Talbert, 2025). Her experience illustrates the lasting value of CPR education and the importance of maintaining preparedness for life-threatening emergencies (Ng et al., 2023). CPR education has increasingly been recognized as an essential component of emergency preparedness, while community-based training has been associated with improved survival outcomes after out-of-hospital cardiac arrest (Ng et al., 2023; Simmons et al., 2023). International resuscitation organizations likewise promote CPR and Basic Life Support (BLS) training to expand public preparedness and the pool of capable first responders (Wyckoff et al., 2022). As access to CPR training has expanded, attention has also turned to how well that training translates into actual performance, leading to growing interest in technologies that can objectively measure and improve compression quality (Ma et al., 2025; Nicolau et al., 2024; Kahsay et al.,

2024; Gharibi et al., 2026; Leo et al., 2023; Kwon et al., 2022). While these efforts have largely expanded on a global scale, the Philippines is also beginning to strengthen its own CPR education, competency, and performance monitoring.

The growing emphasis on emergency preparedness has been accompanied by the growing integration of CPR education across government agencies, communities, and schools. The Department of Health (DOH) continues to encourage Filipinos to acquire CPR knowledge and skills to enable immediate responses during emergencies (Montemayor, 2024). Similarly, the Philippine Red Cross has strengthened public awareness and practical competence through initiatives such as the ASAP strategy—Awareness, Skills, Access, and Perform—which promotes wider participation in CPR education and training and includes accessible approaches such as Do-it-Yourself (DIY) CPR mannequins for community-based BLS initiatives (Philippine Red Cross [PRC], 2025). This national emphasis is likewise reflected in Bataan through government- and school-based initiatives: the Provincial Disaster Risk

Reduction and Management Office (PDRMO) conducted hands-only CPR and Automated External Defibrillator (AED) training for Provincial Government employees, while Bataan National High School-Senior High School (BNHS-SHS) conducted CPR Awareness Training in partnership with the Philippine Red Cross (Marcelo & Marcelo, 2022). Records from BNHS-SHS further indicate that more than 1,000 STEM students participated in CPR training activities during the first five years of Senior High School implementation. The integration of Basic Life Support (BLS) training into the Strengthened Senior High School curriculum for Grade 12 learners in STEM-Health Allied, Sports, and Pre-Baccalaureate Maritime (PBM) further demonstrates this emphasis within schools. Taken together, these developments highlight the growing integration of CPR education across Philippine and local educational settings and provide a context for exploring how technology-assisted approaches may complement existing CPR training practices.

High-quality cardiopulmonary resuscitation (CPR) is determined not only by whether chest compressions are performed but also by how accurately they conform to established standards of CPR quality. Current American Heart Association guidelines identify compression depth, compression rate, complete chest recoil, and chest compression fraction (CCF) as fundamental indicators of effective chest compressions. Adult CPR should be delivered at a rate of 100–120 compressions per minute and a depth of at least 5 cm

while avoiding depths greater than 6 cm (Kleinman et al., 2025). The guidelines further emphasize complete chest recoil between compressions and minimizing interruptions to maintain an adequate CCF. These indicators provide objective criteria for evaluating the quality and continuity of CPR performance. Higher CCF levels, for example, have been associated with improved rates of return of spontaneous circulation and survival among cardiac arrest patients (Leo et al., 2023; Maddani et al., 2022; Yang et al., 2025).

The existence of established standards does not necessarily ensure that they are consistently achieved during training. Compression rate can be quantified by measuring the number of compressions delivered within a specified period, whereas compression depth requires accurate measurement of chest displacement during each compression (Rao et al., 2025; Stumpf et al., 2022). Evaluation of chest recoil requires confirmation that the chest fully returns to its original position between compressions (de Borba et al., 2026; Wingen et al., 2024), while determining CCF requires monitoring the continuity of chest compressions throughout a resuscitation attempt (Maddani et al., 2022). Assessing these indicators therefore requires accurate measurement.

Existing research has established both the educational value and technical feasibility of technology-assisted CPR feedback, with studies reporting improved CPR performance and demonstrating the capacity of sensor-based systems,

smartphone-assisted applications, and feedback-enabled manikins to monitor CPR quality indicators during training (Lin et al., 2025; Nicolau et al., 2024; Chang et al., 2023; Lee et al., 2023; Vallianatou et al., 2025). However, existing systems vary in learner populations, instructional settings, technical configurations, and implementation approaches, indicating that their applicability may differ across educational contexts.

Based on the preliminary interviews conducted with CPR instructors from BNHS-SHS, the Balanga City Disaster Risk Reduction and Management Office (DRRMO), the Philippine Red Cross-Bataan Chapter, and the Bureau of Fire Protection in Region III (Balanga City), instructor observation, return demonstrations, and competency checklists remain integral to CPR education but become more limited when multiple CPR quality indicators must be assessed simultaneously. A comparison between instructor assessments and device-generated measurements reported only moderate-to-low agreement in evaluating compression depth, compression rate, and chest recoil (Lin et al., 2025). The issue is not a lack of instructor expertise; rather, certain CPR quality indicators are inherently difficult to evaluate precisely through observation alone. The interviews also reflected these concerns within the intended training environment. The absence of objective measurement tools was identified as a recurring limitation during CPR practice sessions, resulting in substantial reliance on observational assessment. Findings from a preliminary

questionnaire conducted among students who had previously experienced BLS/CPR training at BNHS-SHS further corroborated these accounts: among the 20 respondents who reported receiving school-based CPR training, 15 indicated that they practiced primarily on basic manikins without performance feedback, and feedback was most frequently delivered through instructor observation or verbal feedback. Difficulties in determining whether compressions consistently achieved the recommended depth, rate, and recoil standards were also noted, with 11 respondents reporting that they were sometimes unsure whether they were performing CPR according to the recommended technique and 2 reporting that they were often unsure. Such conditions may reduce opportunities for the immediate identification and correction of performance errors, particularly when assessment depends primarily on visual judgment.

Group-based instruction introduces an additional challenge because instructors may need to supervise several learners simultaneously, limiting continuous and individualized monitoring. Interview findings indicated that immediate individualized feedback may be affected when several learners perform CPR simultaneously, while the same questionnaire showed that 13 of 20 respondents reported instructor observation only in some instances and 3 reported it rarely. These considerations place measurement performance at the center of the present investigation and establish the basis for examining a related concern: system functionality.

The delivery of real-time feedback requires efficient communication between the sensing component and the application interface, where performance information must be transmitted both promptly and reliably. Timely data transmission is essential to maintain synchronization between learner actions and system responses, as delays in communication may reduce system responsiveness and the usefulness of the feedback provided. Research on real-time biofeedback systems has emphasized the importance of timely data transmission, noting that latency and communication delays can diminish system responsiveness and user interaction (Kos et al., 2025). Beyond transmission speed, communication must remain reliable to ensure performance data are delivered consistently and completely from the sensing device to the application. Similarly, studies on healthcare monitoring systems have identified interruptions in wireless communication and packet loss as factors that may affect the integrity and completeness of transmitted information (Franco et al., 2026; Verma et al., 2025). Thus, effective real-time feedback depends on communication that is both timely and reliable, maintaining synchronization with learner actions while preserving the integrity of transmitted performance information.

Beyond immediate feedback, performance records may allow learners and instructors to review previous sessions and monitor progress, as demonstrated by smartphone-assisted CPR applications (Lee et al., 2023; Vallianatou et al., 2025). This is particularly relevant because CPR skills may

decline without continued reinforcement (Onbasilar et al., 2025), although long-term retention is beyond the scope of the present study. The same questionnaire indicated that 12 of 20 respondents had no performance record to review after practice, while 8 reported that their results were not recorded or saved.

System functionality also depends on the correct interpretation of measured values, as sensor readings must be processed according to appropriate criteria before meaningful feedback can be generated. Research indicates that estimated compression depth and rate depend not only on sensors but also on the algorithms used to process and interpret their data (de Borja et al., 2026; Gould et al., 2025; Rao et al., 2025). These requirements are relevant within the intended educational setting, where interview and questionnaire findings identified limitations in training resources and shared equipment: while 17 respondents practiced on DIY CPR manikins, 2 reported that no manikin was provided and 1 described using a stuffed toy instead. Such constraints may limit consistent monitoring and repeated practice, while equipment and economic barriers have also been identified in educational CPR programs (Peixoto-Pino et al., 2024). These constraints do not independently establish the need for a feedback system; however, they indicate a context where supplementary technology may support performance assessment, feedback delivery, and training documentation within existing arrangements.

Across existing technology-assisted CPR systems, implementation varies in learner populations, instructional settings, and technological configurations. Findings from the interviews and questionnaire similarly indicate continued reliance on observational assessment, limited opportunities for individualized monitoring, and gaps in access to training resources and performance records.

Guided by these gaps, this study aims to develop and evaluate CPReady, an ESP32-based supplementary cardiopulmonary resuscitation (CPR) training device and mobile application designed to provide objective monitoring of CPR performance and real-time feedback during practice. It seeks to describe the accuracy of the prototype's CPR performance measurements and evaluate the functionality of its integrated sensing, wireless communication, data logging, and firmware algorithmic accuracy.

This study aims to contribute to school-based CPR education by developing CPReady, a supplementary device for instructor-led CPR training amongst STEM-Health Allied students that integrates real-time feedback for the important CPR indicators set by the American Heart Association—such as compression depth, compression rate, chest compression fraction, and chest recoil—with added support from a mobile application with progress tracking, guided learning, and scenario-based learning features. Under these circumstances, the study will provide evidence regarding the accuracy of the

aforementioned CPR performance measurements and the integrated system functionalities. Furthermore, the development of the proposed device may contribute to groups such as the CPR instructors and trainers, STEM Health Allied programs, and future research.

OBJECTIVE OF THE STUDY

The study aims to develop and evaluate CPReady, an ESP32-based supplementary cardiopulmonary resuscitation (CPR) training device and mobile application, and describe its CPR performance measurement accuracy and system functionality. Specifically, it seeks to describe:

- Compression depth;
- Compression rate;
- Chest recoil detection;
- Chest Compression Fraction (CCF);
- Response time from sensor detection to mobile application display;
- Bluetooth data transmission reliability;
- Data logging completeness; and
- Firmware algorithmic accuracy.

STATEMENT OF THE PROBLEM/RESEARCH QUESTIONS

The researchers aim to develop and evaluate CPReady, an ESP32-based supplementary CPR training device and mobile application, and describe the accuracy of its CPR performance measurements and functionality.

Specifically, the study seeks to answer the following research questions:

1. How **accurate** are the CPR performance measurements of the developed CPReady prototype in terms of:
 - 1.1 Compression depth (cm);
 - 1.2 Compression rate (compressions per minute);
 - 1.3 Chest recoil detection; and
 - 1.4 Chest Compression Fraction (%)?
2. How may the system functionality of the developed CPReady prototype be described in terms of:
 - 2.1 Response time from sensor detection to mobile application display (seconds);
 - 2.2 Bluetooth data transmission reliability (%);
 - 2.3 Data logging completeness (%); and
 - 2.4 Firmware algorithmic accuracy (%).

MATERIALS AND METHODS

METHODOLOGY AND DESIGN

The study will employ a quantitative research approach using a descriptive research design to evaluate the measurement accuracy and system functionality of the developed Cardiopulmonary Resuscitation (CPR) supplementary training device. The quantitative approach will allow the performance of the prototype to be examined through numerical measurements obtained during prototype testing. Within this

approach, the descriptive design will be used to characterize the prototype's performance based on the collected measurements. This application is consistent with Slater and Hasson (2024), who describe quantitative research as centered on the systematic collection and analysis of numerical data and descriptive designs as structured approaches that provide a clear picture of the subject being investigated.

The descriptive evaluation will focus on two domains: measurement accuracy and system functionality. Measurement accuracy will be evaluated for compression depth, compression rate, chest recoil detection, and Chest Compression Fraction (CCF) by examining the prototype's outputs in relation to the corresponding reference measurements or classifications. System functionality will be evaluated in terms of response time from sensor detection to mobile application display, Bluetooth data transmission reliability, data logging completeness, and firmware algorithmic accuracy. The numerical results obtained from prototype testing will then be summarized and analyzed using descriptive statistics to characterize the prototype's performance across these measures.

INSTRUMENT

To collect the quantitative data necessary to evaluate the measurement accuracy and system functionality of the developed CPR supplementary training device, the study will **utilize an existing CPR feedback or measurement system**, video recording and manual timing, and a

researcher-made prototype testing data collection sheet. In addition, a certified Basic Life Support (BLS) instructor will assist in verifying reference classifications that require expert assessment. The CPR performance criteria and expected classifications used during the evaluation will be based on the standards adopted by the study—the 2025 American Heart Association (AHA) Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care.

The existing CPR feedback or measurement system will serve as a comparison instrument for evaluating the prototype's measurement accuracy in compression depth and compression rate. During prototype testing, measurements obtained from the selected system will be recorded alongside the corresponding measurements generated by the prototype, providing a comparable basis for evaluating the accuracy of its outputs. Because video recording allows time-sensitive CPR events, including compression periods and interruptions, to be reviewed retrospectively (Wolfsohn et al., 2026), video recording and manual timing will be used to provide an independent time reference for determining CCF and response time. The recorded trials will be used to determine the duration of active chest compressions in relation to the total CPR session for CCF and to measure response time based on the interval between a compression event and its corresponding display on the mobile application. A certified BLS instructor will verify the reference

classification for chest recoil detection as complete or incomplete.

A researcher-made prototype testing data collection sheet will be used to systematically record and consolidate the measurements and results obtained throughout prototype testing. It will document the comparison measurements or verified classifications together with their corresponding prototype outputs, as well as the data obtained from the functionality tests. These will include response-time measurements, transmitted and received Bluetooth packets, expected and successfully stored records, and expected and actual firmware classifications, which will provide the data required to evaluate response time, Bluetooth data transmission reliability, data logging completeness, and firmware algorithmic accuracy.

The collected data will be analyzed using mean and standard deviation (SD) to summarize numerical measurements and their variability. Mean difference, mean absolute error (MAE), and percentage error will be used to quantify differences between the prototype and corresponding reference measurements for compression depth, compression rate, and CCF. Classification-based results, including chest recoil detection and firmware algorithmic accuracy, will be summarized using frequency and percentage of correct classifications. Response time will be described using the mean, minimum, maximum, and SD. Bluetooth data transmission reliability, data logging completeness, and firmware algorithmic accuracy will be expressed as percentages

based on successfully received packets, successfully stored records, and correctly classified outputs, respectively. These descriptive measures will be considered together in evaluating the measurement accuracy and system functionality of the prototype.

ETHICAL CONSIDERATIONS

According to the University of Pennsylvania (2024), ethical considerations in research consist of principles that would guide the study. These principles aim to protect the rights and welfare of participants while ensuring the credibility, transparency, and integrity of research findings. The primary ethical concerns in this study involve ensuring the accuracy and consistency of CPR performance measured by the CPReady prototype, maintaining the confidentiality of respondents, and minimizing researcher bias during data collection and analysis.

To address these concerns, the researchers will follow the ethical principles stated by the University of Pennsylvania (2024). The accuracy and reliability of the CPReady prototype will be supported with appropriate development procedures, standardized testing tools, and repeated trials to verify the consistency of its measurements and functions. To minimize bias and maintain objectivity, the researchers will keep complete and accurate records of all collected data, acknowledge the limitations and present findings encountered during the study. Furthermore, prototype evaluations and research

procedures will be reviewed through consultation with the research adviser and relevant experts to ensure the credibility and trustworthiness of the research.

Ethical considerations regarding the participation of students, teachers, and CPR experts will be observed throughout the study. All participants will be provided with an informed consent form regarding the purpose of the research, the procedures involved, potential risks and benefits, and their right to withdraw from the study at any time without consequences. The researchers will protect the privacy and confidentiality of participants by securing collected data and ensuring that personal identities are not disclosed without permission. Moreover, participants will be treated with respect and fairness, and no unnecessary physical, emotional, or psychological harm will be inflicted during the conduct of the study. Since CPReady is intended solely as an educational CPR training prototype, participants will also be informed that it is not a medical device and is not intended for actual patient care or emergency response situations. These measures aim to safeguard the rights, privacy, and welfare of all participants while ensuring that the study is conducted responsibly and ethically.

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